AlyahHub POS & Inventory System

Complete API Documentation & Integration Specification

Version 1.0 · Built on Laravel 13 · REST/JSON · Sanctum Bearer Authentication

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1. Project Overview & Core Architecture

AlyahHub is an enterprise-grade **mobile phone shop inventory management and Point of Sale (POS) system** built on **Laravel 13**. It provides a complete multi-tenant, role-aware RESTful API serving retail cashiers, store managers, and system administrators.

Core System Features

Feature	Description
Product Catalog & Variant Management	Device hardware attributes: storage capacity, colorways, IMEI/barcode, unit prices, wholesale tiers
Multi-Store & Stock Location Tracking	Warehouse, backroom, and shop-floor shelf bin management across multiple retail branches
Auditable Inventory Ledger	Strict stock movement tracking across purchase orders, sales checkouts, adjustments, and returns
Procurement & Supplier Workflows	Supplier records, purchase orders, goods receipts, and payment allocations
POS & Cash Shift Management	Cash registers, cashier shift drawer open/close, real-time sales, cash discrepancy auditing
Reporting & Analytics (Phase 6)	Real-time operational queries and historical aggregations for revenue, reorder alerts, and shift reconciliation

Core Inventory Principle

> **Immutable Ledger Enforcement:** Stock movements **MUST** be recorded through the inventory transaction ledger (`stock_movements` table). The system **MUST NOT** directly overwrite stock balances without creating an auditable ledger movement entry.

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2. Base URL & Common Headers

Base URL

During local development, all API routes are exposed under the `/api` prefix:

```
http://127.0.0.1:8000/api
```

Request Headers

Header	Value	Required
`Content-Type`	`application/json`	Yes
`Accept`	`application/json`	Yes
`Authorization`	`Bearer {{YOUR_ACCESS_TOKEN}}`	Protected routes only

Standard Success Response

```
{
  "message": "Resource retrieved successfully",
  "data": {}
}
```

Validation Error Response (HTTP 422)

```
{
  "message": "The given data was invalid.",
  "errors": {
    "email": ["The email field is required."],
    "password": ["The password field must be at least 8 characters."]
  }
}
```

3. Roles & Permissions Matrix

AlyahHub uses **Spatie Laravel-Permission** for fine-grained authorization. Role and permission checks are evaluated inside controller logic or via standard Sanctum middleware.

Module / Action	Admin	Manager	Sales / Cashier
Manage Users & Roles	✅	❌	❌
View Catalog (Categories, Brands, Products)	✅	✅	✅
Create / Edit / Delete Catalog Items	✅	✅	❌
Manage Stores & Stock Locations	✅	✅	❌
Create Purchase Orders & Receive Goods	✅	✅	❌
Open / Close POS Session Shift	✅	✅	✅
Process POS Sales Checkout	✅	✅	✅
View System Analytics & Reports	✅	✅	❌

4. API Reference

Endpoint Index

Module	Method	Endpoint	Access
Auth	POST	`/api/register`	Public (dev) / Admin (prod)
Auth	POST	`/api/login`	Public
Auth	GET	`/api/me`	Bearer token
Auth	POST	`/api/logout`	Bearer token
Catalog	GET/POST/PUT/PATCH/DELETE	`/api/categories`	Catalog roles
Catalog	GET/POST/PUT/DELETE	`/api/brands`	Catalog roles
Catalog	GET/POST/PUT/DELETE	`/api/units`	Catalog roles
Catalog	GET/POST/PUT/DELETE	`/api/products`	Catalog roles

Module	Method	Endpoint	Access
Catalog	GET/POST/PUT/PATCH/DELETE	`/api/product-variants`	Catalog roles
Organization	GET/POST	`/api/stores`	Admin, Manager
Organization	GET/POST	`/api/stock-locations`	Admin, Manager
Inventory	GET	`/api/stock-balances`	Authenticated
POS	POST	`/api/registers`	Admin, Manager
POS	POST	`/api/pos-sessions`	All roles
POS	POST	`/api/pos-sessions/{id}/close`	All roles
POS	POST	`/api/sales`	All roles
Reports	GET	`/api/reports/sales`	Admin, Manager
Reports	GET	`/api/reports/low-stock`	Admin, Manager
Reports	GET	`/api/reports/cash-reconciliation`	Admin, Manager

Module 1: Authentication & Identity Management

1.1 Register New User

`POST /api/register`

Access: Public (Development Mode) / Admin Restricted (Production Mode) **Model** **`$fillable`**: `name`, `email`, `password`

Request Body

```
{
  "name": "Dawit Tadesse",
  "email": "dawit@alyahhub.com",
  "password": "Password123!",
  "password_confirmation": "Password123!",
  "role": "sales"
}
```

Response — 201 Created

```
{
  "message": "User registered successfully",
  "token": "1|qX8zL0pQv...",
  "user": {
    "id": 1,
    "name": "Dawit Tadesse",
    "email": "dawit@alyahhub.com",
    "roles": ["sales"],
    "created_at": "2026-10-24T01:00:00.000000Z"
  }
}
```

1.2 User Login

`POST /api/login` · **Access:** Public

Request Body

```
{
  "email": "dawit@alyahhub.com",
  "password": "Password123!"
}
```

Response — 200 OK

```
{
  "message": "Login successful",
  "token": "2|yR9aM1qR...",
  "user": {
    "id": 1,
    "name": "Dawit Tadesse",
    "email": "dawit@alyahhub.com",
    "roles": ["sales"]
  }
}
```

1.3 Get Current Authenticated User

`GET /api/me` · **Access:** Protected (Bearer Token)

Response — 200 OK

```
{
  "message": "Authenticated user profile retrieved",
  "user": {
    "id": 1,
    "name": "Dawit Tadesse",
    "email": "dawit@alyahhub.com",
    "roles": ["sales"],
    "permissions": ["process sales", "open pos session"]
  }
}
```

1.4 User Logout

`POST /api/logout` · **Access:** Protected (Bearer Token)

Response — 200 OK

```
{
  "message": "Token revoked successfully"
}
```

Module 2: Product Catalog & Master Data

2.1 Categories

Method	Endpoint	Purpose
GET	`/api/categories`	List all active categories
POST	`/api/categories`	Create category
GET	`/api/categories/{id}`	Retrieve category details
PUT/PATCH	`/api/categories/{id}`	Update category fields
DELETE	`/api/categories/{id}`	Soft delete or remove category

Model `$fillable`: `name`, `code`, `description`, `parent_id`, `is_active`

Create Request — `POST /api/categories`

```
{
  "name": "Smartphones",
  "code": "CAT-SMARTPHONE",
  "description": "Mobile phones and handheld smart devices",
  "is_active": true
}
```

2.2 Brands

Endpoints: `GET`, `POST`, `GET /{id}`, `PUT /{id}`, `DELETE /{id}` on `/api/brands` **Model** **`$fillable`:** `name`, `code`, `is_active`

```
{
  "name": "Apple",
  "code": "BRAND-APPLE",
  "is_active": true
}
```

2.3 Measurement Units

Endpoints: `GET`, `POST`, `GET /{id}`, `PUT /{id}`, `DELETE /{id}` on `/api/units` **Model** **`$fillable`:** `name`, `code`, `is_active`

```
{
  "name": "Piece",
  "code": "PCS",
  "is_active": true
}
```

2.4 Master Products

Endpoints: `GET`, `POST`, `GET /{id}`, `PUT /{id}`, `DELETE /{id}` on `/api/products`

Model **`$fillable`:** `category_id`, `brand_id`, `base_unit_id`, `product_code`, `name`, `slug`, `description`, `status`, `track_inventory`, `is_sellable`, `created_by`

```
{
  "category_id": 1,
  "brand_id": 1,
  "base_unit_id": 1,
  "product_code": "PROD-IP15P",
  "name": "iPhone 15 Pro",
  "slug": "iphone-15-pro",
  "description": "Apple iPhone 15 Pro Titanium Edition",
  "status": "active",
  "track_inventory": true,
  "is_sellable": true
}
```

2.5 Product Variants

Method	Endpoint
GET	`/api/products/{productId}/variants`
POST	`/api/product-variants`
GET	`/api/product-variants/{id}`
PUT/PATCH	`/api/product-variants/{id}`
DELETE	`/api/product-variants/{id}`

Model `$fillable`: `product_id`, `sku`, `barcode`, `variant_name`, `attributes`, `cost_price`, `selling_price`, `wholesale_price`, `minimum_selling_price`, `weight`, `is_active`

Create Request — `POST /api/product-variants`

```
{
  "product_id": 1,
  "sku": "IP15P-256GB-BLK",
  "barcode": "194253000000",
  "variant_name": "iPhone 15 Pro - 256GB Black Titanium",
  "attributes": {
    "storage": "256GB",
    "color": "Black Titanium"
  },
  "cost_price": 999.00,
  "selling_price": 1199.00,
  "wholesale_price": 1099.00,
  "minimum_selling_price": 1050.00,
  "weight": 0.187,
  "is_active": true
}
```

Module 3: Organization & Inventory Management

3.1 Stores

Endpoints: `POST /api/stores`, `GET /api/stores` **Model** **`$fillable`:** `code`, `name`, `phone`, `email`, `address`, `is_active`

3.2 Stock Locations

Endpoints: `POST /api/stock-locations`, `GET /api/stock-locations` **Model** **`$fillable`:** `store_id`, `code`, `name`, `location_type`, `is_active`

3.3 Stock Balances & Inventory Ledger

`GET /api/stock-balances`

Monitors physical `quantity_on_hand`, `quantity_reserved`, and `reorder_level` per variant per stock location.

Module 4: POS & Sales Checkout Workflow

4.1 Create Register

`POST /api/registers` **Model** **`$fillable`:** `register_id`-scoped fields: `store_id`, `name`, `code`, `is_active`

4.2 Open POS Session (Cashier Shift Start)

`POST /api/pos-sessions` · **Access:** Cashier / Sales token **Model** **`$fillable`**: `register_id`, `user_id`, `opening_balance`, `status`, `opened_at`, `notes`

```
{
  "register_id": 1,
  "opening_balance": 1000.00,
  "notes": "Morning cashier shift startup"
}
```

4.3 Process Sale Checkout

`POST /api/sales` · **Access:** Cashier / Sales token **Model** **`$fillable`**: `pos_session_id`, `customer_id`, `invoice_number`, `discount_total`, `tax_total`, `grand_total`, `notes`

```
{
  "pos_session_id": 1,
  "customer_id": 1,
  "invoice_number": "INV-2026-0001",
  "discount_total": 0.00,
  "tax_total": 0.00,
  "notes": "Retail walk-in transaction",
  "items": [
    {
      "product_variant_id": 1,
      "quantity": 1.000,
      "unit_price": 1199.00,
      "discount": 0.00
    }
  ],
  "payments": [
    {
      "payment_method": "cash",
      "amount": 1200.00,
      "reference_number": null
    }
  ]
}
```

4.4 Close POS Session (Shift Audit)

`POST /api/pos-sessions/{id}/close`

```
{
  "closing_balance": 2199.00,
  "notes": "Shift ended. Register balanced with total drawer cash."
}
```

Module 5: Reporting & Analytics (Phase 6)

All report endpoints require `Authorization: Bearer {{admin_token}}` with the *view reports* permission.

5.1 Sales Summary & Revenue Report

`GET /api/reports/sales`

Parameter	Format	Required
`start_date`	`YYYY-MM-DD`	Yes
`end_date`	`YYYY-MM-DD`	Yes
`store_id`	integer	No

Example: `http://127.0.0.1:8000/api/reports/sales?start_date=2026-10-01&end_date=2026-10-31&store_id=1`

Response — 200 OK

```
{
  "message": "Sales report retrieved successfully",
  "data": {
    "total_sales_count": 1,
    "total_revenue": "1199.00",
    "total_discount": "0.00",
    "total_tax": "0.00",
    "sales": [
      {
        "id": 1,
        "invoice_number": "INV-2026-0001",
        "grand_total": "1199.00",
        "created_at": "2026-10-24T01:10:00.000000Z"
      }
    ]
  }
}
```

5.2 Low Stock Inventory Alert Report

`GET /api/reports/low-stock`

Parameter	Format	Required
`store_id`	integer	No

Example: `http://127.0.0.1:8000/api/reports/low-stock?store_id=1`

Response — 200 OK


```
{
  "message": "Low stock report retrieved successfully",
  "data": [
    {
      "id": 1,
      "stock_location_id": 1,
      "product_variant_id": 1,
      "quantity_on_hand": 2,
      "reorder_level": 5,
      "product_variant": {
        "sku": "IP15P-256GB-BLK",
        "variant_name": "iPhone 15 Pro - 256GB Black"
      }
    }
  ]
}
```


5.3 Cash Drawer Reconciliation Report

`GET /api/reports/cash-reconciliation`

Parameter	Format	Required
`register_id`	integer	No

Example: `http://127.0.0.1:8000/api/reports/cash-reconciliation`

Response — 200 OK

```
{
  "message": "Cash drawer reconciliation report retrieved successfully",
  "data": [
    {
      "session_id": 1,
      "register": "Checkout Counter 1",
      "store": "Bole Central Store",
      "cashier": "Dawit Tadesse",
      "opening_balance": "1000.00",
      "cash_sales_total": "1199.00",
      "expected_cash_balance": "2199.00",
      "closing_balance": "2199.00",
      "discrepancy": 0.00,
      "opened_at": "2026-10-24 08:00:00",
      "closed_at": "2026-10-24 17:00:00",
      "notes": "Shift ended. Register balanced with total drawer cash."
    }
  ]
}
```


5.4 Pre-Aggregated Daily Sales Summary (`DailySalesSummary`)

Model `$fillable`: `store_id`, `report_date`, `total_sales_count`, `gross_revenue`, `total_discount`, `total_tax`

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5. HTTP Status Code Reference

Code	Status	Description & Usage Context
200	OK	Standard successful response for GET, PUT, or general queries
201	Created	A new entity (user, category, product, sale) was created
400	Bad Request	Malformed JSON payload or invalid parameters
401	Unauthorized	Bearer token is missing, expired, or invalid
403	Forbidden	User lacks the required Spatie role permission for the endpoint
404	Not Found	Route, endpoint, or resource ID does not exist

Code	Status	Description & Usage Context
422	Unprocessable Entity	Validation failed (e.g., duplicate email, missing field)
500	Internal Server Error	Unexpected server-side exception or unhandled database constraint